CERTIFICATE OF INSTALLATION

Note: This table completed by ~~HERS~~ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. General Information

Notes:

- The outdoor design temperatures for heating shall be ≥99.0% Heating Dry Bulb or the Heating Winter Median of Extremes values.
- The outdoor design temperatures for cooling shall be ≤1.0% Cooling Dry Bulb and Mean Coincident Wet Bulb values.

01	Dwelling Unit Name		02	Climate Zone	
03	Dwelling Unit Total Conditioned Floor Area (ft^2)		04	Number of Space Conditioning Systems in this Dwelling Unit	
05	Certificate of Compliance Type		06	Method Used to Calculate HVAC Loads (See Section 150.0(h))	
07	Outdoor Design Condition Source (See Section 150.0(h)2		08	Cooling Outdoor Design Temperature Selected (°F)	
09 ~~7~~	~~Calculated Dwelling Unit Sensible Cooling Load (Btu/h)~~ Heating Outdoor Design Temperature Selected (°F)		10 ~~08~~	~~Calculated Dwelling Unit Heating Load (Btu/h)~~ Calculated Dwelling Unit Sensible Cooling Load (Btu/h)	
11 ~~09~~	~~Dwelling Unit Number of Bedrooms~~ Calculated Dwelling Unit Heating Load (Btu/h)		12	Dwelling Unit Number of Bedrooms	

MCH-01c - Space Conditioning Systems Ducts and Fans - Prescriptive, Newly Constructed Buildings

B. Design Space Conditioning (SC) System Component Specifications from CF1R

This table reports the space conditioning system features that were specified on the registered CF1R compliance document for this project.

01	02	03	04	05	06	07	07b	08	09	10	11	12
SC System ID/Name from CF1R	Heating System Type	Heating Efficiency Type	Heating Efficiency Value	Cooling System Type	Cooling Efficiency Type	Cooling Efficiency Value SEER/SEER2	Cooling Efficiency Value EER/EER2/ CEER	Distribution System Type	Duct Location	Duct R-value	Thermostat Type	Comments

C. Installed Space Conditioning (SC) System Component Information

01	02	03	04	05	06	07	08	09	10	11
SC System ID/Name from CF1R	SC System Description of Area Served	Conditioned Floor Area Served by the System (ft^2)	Heating System Type	Cooling System Type	Distribution System Type	Duct Location	SC System Thermostat Type	Cooling Zoning Type	Cooling System Compressor Speed Type	Number of Indoor Units for this System
Notes:										

D. Installed Heating Equipment Information (not heat pumps)

01	02	03	04	05	06	07	08	09	00	11
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Does Indoor Unit Provide CFI IAQ Ventilation?	Indoor Unit Duct Status	Heating Efficiency Type	Heating Efficiency Value	Heating Unit Manufacturer	Heating Unit Model Number	Heating Unit Serial Number	Rated Heating Capacity Output (Btu/h)
Notes:										

E. Installed Cooling System Outdoor Condensing Unit or Package Unit Equipment Information (not heat pumps)

01	02	03	04	04b	05	06	07	08	09
SC System ID/Name from CF1R	SC System Description of Area Served	Cooling Efficiency Type	Cooling Efficiency Value SEER/SEER2	Cooling Efficiency Value EER/EER2/CEER	Condenser or Package Unit Manufacturer	Condenser or Package Unit Model Number	Condenser or Package Unit Serial Number	System Cooling Capacity at Design Conditions (Btu/h)	Condenser Nominal Cooling Capacity (ton)
Notes:									

F. Installed Split System Indoor Unit (Coil or Fan Coil) Equipment Information - applicable to DX or hydronic, heating or cooling, coils and fan coil units.

Systems with more than one indoor coil or fan coil unit (e.g. multi-split systems) shall provide information for each of the system indoor unit coils or fan coil units.

01	02	03	04	05	06	07	08	09	10
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Indoor Unit Type	Indoor Unit Duct Status	Does Indoor Unit Provide CFI IAQ Ventilation?	Indoor Unit Manufacturer	Indoor Unit Model Number	Indoor Unit Serial Number	Indoor Unit Nominal Cooling Capacity (ton)
Notes:									

G. Installed Heat Pump System – Split System Condensing Unit or Package Unit Equipment Information

01	02	03	04	05
SC System ID/Name from CF1R	SC System Description of Area Served	Condenser or Package Unit Manufacturer	Condenser or Package Unit Model Number	Condenser or Package Unit Serial Number
Notes:				

H. Installed Heat Pump System – Efficiency and Performance Compliance Information

01	02	03	04	05	06	07	08	08b	09	10
SC System ID/Name from CF1R	SC System Description of Area Served	Heating Efficiency Type	Heating Efficiency Value	System Rated Heating Capacity at 47°F (Btu/h)	System Rated Heating Capacity at 17°F (Btu/h)	System Cooling Efficiency Type	System Rated Cooling Efficiency Value SEER/SEER2	System Rated Cooling Efficiency Value EER/EER2/CEER	System Cooling Capacity at Design Conditions (Btu/h)	Condenser Nominal Cooling Capacity (ton)
Notes:										

I. Installed Duct System Information

01	02	03	04	05	06	07	08	09	10	11	12	13	14
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Supply Duct Location	Supply Duct R-Value	Return Duct Location	Return Duct R-Value	Exception from Min R-Value for Ducts In Conditioned Space	Method of compliance with Airflow and Fan Efficacy Req's in 150.0(m)13	Number of Air Filter Devices on System	Can Approved Airflow Protocols be used to test this System?	Can Approved Fan Efficacy Protocol be used to test this System?	Total Duct Length	Required New Duct R-Value
Notes:													

J. Installed Air Filter Device Information

Mandatory requirements for air filter devices are specified Section 150.0(m)12. The installer shall place a sticker in or near each filter grille that displays the design airflow rate for that filter grille/rack and the maximum allowed clean filter pressure drop at the design airflow rate. This will inform the occupant of the airflow vs pressure drop performance required for replacement air filters.

01	02	03	04	05	06	07	08	09	10	11	12	013
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Air Filter Name or Description of Location	Air Filter Rack Type	Design Airflow Rate for Air Filter Device (cfm)	Air Filter Nominal Depth (inch)	Air Filter Nominal Length (inch)	Air Filter Nominal Width (inch)	Air Filter Calculated Nominal Face Area ($inch^2$)	Air Filter Required Minimum Face Area ($inch^2$)	Face Area Compliance	Design Allowable Pressure Drop for Air Filter Device (inch W.C.)
Notes:												

K. Air Filter Device Requirements

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

Mandatory Air Filter Device Requirements can be found in Section 150.0(m)12A-E. Some mandatory requirements may apply in addition to those listed below.

01	All recirculated air and all outdoor air (including make up air) supplied to the occupiable space is filtered before passing through the system's thermal conditioning components.
02	The space conditioning system shall be designed to accommodate the clean-filter pressure drop imposed by the system air filter device(s). The design airflow rate and maximum allowable clean-filter pressure drop at the design airflow rate applicable to each air filter device shall be determined by the system designer. The system installer shall affix a sticker/label to each system air filter grille/rack locations that discloses the filter's design airflow rate and the filter's maximum allowable clean-filter pressure drop at the design airflow rate. The sticker/label shall be permanently affixed to the air filter device, readily legible, and visible to a person replacing the air filter.
03	All system air filters shall be located and installed in such a manner as to allow access and regular service by the system owner.
04	The system shall be provided with air filter media having a designated efficiency equal to or greater than MERV 13 when tested in accordance with ASHRAE Standard 52.2, or a particle size efficiency rating equal to or greater than 50% in the 0.30-1.0 µm range and equal to or greater than 85 percent in the 1.0-3.0 µm range when tested in accordance with AHRI Standard 680.
05	The system shall be provided with air filters that have been labeled by the manufacturer to disclose efficiency and pressure drop ratings that conform to the efficiency and pressure drop requirements for the air filter grilles/racks.
06	Filter racks or grilles shall use gaskets, sealing, or other means to close gaps around inserted filters and prevent air from bypassing the filter.

L. ~~HERS~~ECC Verification Requirements for Duct Systems

01	02	03	04	05	06	07	09
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	MCH-20 Duct Leakage Test	MCH-21 Duct Location Verification	MCH-22 AHU Fan Efficacy (W/cfm)	MCH-23 AHU Airflow Rate (cfm/ton)	MCH-28 Return Duct Design Table 150.0-B or C
Notes:							

M. ~~HERS~~ECC Verification Requirements for Space Conditioning Equipment

01	02	03
SC System ID or Name from CF1R	SC System Description of Area Served	MCH-25 Refrigerant Charge
Notes:		

N. Space Conditioning Systems, Ducts and Fans – Mandatory Requirements and Additional Measures

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

~~N~~Additional mandatory requirements from Section 150.0 that are not listed here may be applicable to some systems. These requirements may be applicable to only newly installed equipment or portions of the system that are altered. Existing equipment may be exempt from these requirements.

Heating Equipment

01	**Equipment Efficiency:** All heating equipment must meet the minimum efficiency requirements of Section 110.1 and Section 110.2(a) and the Appliance Efficiency Regulations.
02	**Controls:** All unitary heating systems, including heat pumps, must be controlled by a setback thermostat. These thermostats must be capable of allowing the occupant to program the temperature set points for at least four different periods in 24 hours. See Sections 150.0(i), 110.2(b).
03	**Sizing:** Heating load calculations must be done on portions of the building served by new heating systems to prevent inadvertent undersizing or oversizing. See sections 150.0(h)1 and 2~~)~~.
04	**Furnace Temperature Rise:** Central forced-air heating furnace installations must be configured to operate at or below the furnace manufacturer's maximum inlet-to-outlet temperature rise specification. See Section 150.0(h)4.
05	**Standby Losses and Pilot Lights:** Fan-type central furnaces may not have a continuously burning pilot light. Section 110.5 and Section 110.2(d).

Cooling Equipment

06	Equipment Efficiency: All cooling equipment must meet the minimum efficiency requirements of Section 110.1 and Section 110.2(a) and the Appliance Efficiency Regulations.
07	Refrigerant Line Insulation: All refrigerant line insulation in split system air conditioners and heat pumps must meet the R-value and protection requirements of Section 150.0(j)1~~2~~ and 2~~3~~, and Section 150.0(m)9.
08	Condensing Unit Location: Condensing units shall not be placed within 5 feet of a dryer vent outlet. See Section 150.0(h)3A.
09	Liquid Line Filter Drier: A liquid line filter drier shall be installed according to the manufacturer's specifications 150.0(h)3B.
10	Sizing: Cooling load calculations must be done on portions of the building served by new cooling systems to prevent inadvertent undersizing or oversizing. See Section 150.0(h)1 and 2.

Cooling and Heating Equipment (Additional Requirement)

11	**System Selection:** See section 150.0(h)5 A. Equipment sizing and selection shall meet the cooling and heating loads of Section 150.0(h)1 and 2. B. Systems shall be sized based on ACCA Manual S-2023 in accordance with these requirements: i. **Cooling Capacity:** There is no limit on the minimum capacity. ii. **Furnaces**: Heating capacity shall be sized based on ACCA Manual S-2023, Table N2.5. iii. **Heat Pump Heating Capacity**: a. Minimum: Heating systems are required to have a heating capacity meeting the minimum requirements of the CBC not including any supplementary heating. b. Maximum: There is no limit on the maximum heating capacity.
12	**Defrost:** See section 150.0(h)6 and the exceptions. A. If a heat pump is equipped with an installer adjustable defrost delay timer, the delay timer shall be set to greater than or equal to 90 minutes. B. The installer shall certify on the Certificate of Installation (CF2R) that the control configuration has been tested in accordance with the testing procedure in the CF2R. **Exception 1 to Section 150.0(h)6.** Dwelling units in Climate Zones 6 and 7. **Exception 2 to Section 150.0(h)6.** Dwelling units with a conditioned floor area of 500 square feet or less in Climate Zones 3, 5 through 10, and 15.
13	**Supplementary heating control configuration:** See section 150.0(h)7. Heat pumps with supplementary heat, including, but not limited to, electric resistance heaters or gas furnace supplementary heating, shall comply with the following requirements: **See section 150.0(h)7 for exceptions.** A. Lock out supplementary heating above an outdoor air temperature of no greater than 35°F. There are additional thermostat requirements in section 150.0(i)2.

	B. The installer shall certify on the Certificate of Installation that the control configuration has been tested in accordance with the testing procedure found in the CF2R. C. The controls may allow supplementary heater operation above 35°F only during defrost; or when the user selects emergency operation.
14	**Sizing of electric resistance supplementary heat:** See section 150.0(h)8. For heat pumps with electric resistance heating, the capacity of electric resistance heat shall not exceed the heat pump nominal cooling capacity (at 95°F ambient conditions) multiplied by 2.7 kW per ton, rounded up to the closest kW.
15	**Capacity variation with third-party thermostats:** See section 150.0(h)9 Variable or multi-speed systems shall comply with the following requirements: A. The space conditioning system and thermostat together shall be capable of responding to heating and cooling loads by modulating system compressor speed, and meet thermostat requirements in section 150.0(i)2.

Air Distribution System Ducts, Plenums and Fans

~~11~~ 16	Insulation: The minimum duct insulation value is R-6 or ducts can be uninsulated if the duct system is located entirely in conditioned space. Note that higher values may be required by the prescriptive or performance requirements. See Section 150.0(m)1B for exceptions.
~~12~~ 17	Connections and Closures: All installed air-distribution system ducts and plenums must meet the requirements of CMC Sections 601.0, 602.0, 603.0, 604.0, 605.0 and ANSI/SMACNA-006-2006.

Heat Pump Thermostat

~~13~~ 18	A thermostat shall be installed that meets the requirements of Section 110.2(b) and Section 110.2(c).
~~14~~ 19	The thermostat shall be installed in accordance with the manufacturers published installation specifications.
~~15~~ 20	First stage of heating shall be assigned to heat pump heating.
~~16~~ 21	Second stage back up heating shall be set to come on only when the indoor set temperature cannot be met.
22	Setback thermostats: All heating or cooling systems, including heat pumps, not controlled by a central energy management control system (EMCS) shall have a setback thermostat, as specified in Section 110.2(c). See Section 150.0(i)1
23	Thermostats that are applied to heat pumps with supplemental heating: See Section 150.0(i)2 The thermostats controlling heat pumps with electric resistance supplementary heat or gas furnace supplementary heat shall comply with the following requirements: See Section 150.0(i)2 for exceptions. A. The thermostat shall receive outdoor air temperature from an outdoor air temperature sensor or from an internet weather service. B. B. The thermostat shall display the outdoor air temperature. C. The thermostat and heat pump shall lock out supplementary heat when the outdoor air temperature is above 35°F. D. The thermostat shall have an indicator to notify when supplementary heat or emergency heat is in use. E. During defrost or when the user selects emergency heating, supplementary heat operation is permitted above 35°F.

O. Test of Defrost Delay Timer Setting

The installing contractor shall confirm that a heat pump's installer-adjustable Defrost Delay Timer Setting (if it exists) is set to no less than 90 minutes.

01	Test Applicability. Select the statement describing test applicability for this project: 1. The test applies because the heat pump utilizes an installer adjustable Defrost Delay Timer Setting to control defrost and there are no exceptions. 2. The test does not apply because the heat pump does not utilize an installer-adjustable Defrost Delay Timer Setting to control defrost. 3. The test does not apply because Exception 1. Dwelling units in Climate Zones 6 and 7 applies. 4. The test does not apply because Exception 2 . Dwelling units with a conditioned floor area of 500 square feet or less in Climate Zones 3, 5 through 10, and 15 applies.	
02	Recording Configuration of Controls. Specify the mechanism for setting the Defrost Delay Timer Setting (for example, the name of defrost delay timer setting in the thermostat setup, or the location and number of the specific dip switch, jumper, or dial that adjusts Defrost Delay timer).	
03	Record the heat pump's Maximum Available Defrost Delay Timer Setting (minutes).	
04	Record where you set the Defrost Delay Timer Setting (fo example, the numeric timer setting, dip switch position, jumper configuration, or dial setting).	
05	Record where you set the Defrost Delay Timer Setting, in minutes.	
06	Confirming Configuration of Controls. If possible, the Defrost Delay Timer Setting must be 90 minutes or greater. Confirm the Defrost Delay Timer Setting is at least 90 minutes.	

P. Test of Supplementary Heating Lockout

The installing contractor shall confirm the heat pump or thermostat is configured to lock out supplementary heating anytime the outdoor air temperature is above 35°F.

01	Test Applicability. Select the statement describing test applicability for this project: 1. The test applies because the heating system is a heat pump with supplementary heating and there are no exceptions. 2. The test does not apply because the heating system does not include a heat pump with auxiliary heating. 3. The test does not apply because Exception 1. Heat pump being installed in a single family dwelling unit in Climate Zones 7 or 15. 5. The test does not apply because Exception 2 . Heat pump being installed in a single family dwelling unit with a conditioned floor area of 500 square feet or less.	
02	If supplementary heating is electric resistance, what is its rated capacity (kW)?	
03	If supplementary heating is gas, what is its rated input capacity (kBtu/hr)?	
04	Confirming Configuration of Controls. Specify the mechanism for locking out the supplementary heating (for example, the name of supplementary heating lockout setting in the thermostat setup, or the location and number of the specific dip switch, jumper, or dial that adjusts lockout temperature). If there is no mechanism to lock out supplementary heating above a temperature no greater than 35°F this test fails.	
05	Record supplementary heating lock-out setting (for example, the numeric thermostat lock-out setpoint, dip switch position, jumper configuration, or dial setting).	
06	At what Outdoor Air Temperature (OAT) are the controls configured to begin locking out supplementary heating? If this number is above 35°F this test fails.	

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Documentation Author Company Name:	Date Signed:
Address:	CEA/AEA/~~HERS~~ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Installation is true and correct.
2. I am either: a) a responsible person eligible under Division 3 of the Business and Professions Code in the applicable classification to accept responsibility for the system design, construction, or installation of features, materials, components, or manufactured devices for the scope of work identified on this Certificate of Installation, and attest to the declarations in this statement, or b) I am an authorized representative of the responsible person and attest to the declarations in this statement on the responsible person's behalf.
3. The constructed or installed features, materials, components or manufactured devices (the installation) identified on this Certificate of Installation conforms to all applicable codes and regulations and the installation conforms to the requirements given on the Certificate of Compliance, plans, and specifications approved by the enforcement agency.
4. I understand that a registered copy of this Certificate of Installation shall be posted or made available with the building permit(s) issued for the building and shall be made available to the enforcement agency for all applicable inspections. I will take the necessary steps to fulfill~~ensure~~ this requirement~~ is accomplished~~.
5. I understand that a registered copy of this Certificate of Installation is required to be included with the documentation the builder provides to the building owner at occupancy. I will take the necessary steps to fulfill~~ensure~~ this requirement~~ is accomplished~~.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone:	Date Signed:

CF2R-MCH-01c-E User Instructions

Section A. General Information

1. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
2. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
3. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. When the project scope includes an addition to an existing building, the value is equal to the sum of the existing conditioned floor area plus the conditioned floor area of the addition. The default value from the CF1R- may be overwritten in this document. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel
4. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document, but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
5. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
6. Oversized equipment can result in reduced efficiency and capacity. Entirely new systems must be properly sized to match the heating and cooling load of the space that it serves. To do this, heating and cooling load calculations must be performed using an approved calculation methodology. These are listed here. Select the load calculation methodology used for this dwelling unit. If the project consists of a partial replacement of equipment or ducts (change-out), then load calculations are not required. Select N/A. Load calculations are always recommended, especially if the loads of the house have been changed since the original equipment has been installed (reduced via weatherization, other improvements).
7. Enter the Outdoor Design Condition Source (See Section 150.0(h)2), user select from the list.
8. Enter the Cooling Outdoor Design Temperature Selected (°F) for the dwelling unit described by this document.
9. Enter the Heating Outdoor Design Temperature Selected (°F) for the dwelling unit described by this document.

~~7.~~10. Enter the total sensible cooling load for the dwelling unit described by this document. For projects involving dwelling units with more than one system, this will be a sum of the loads for the parts of the dwelling unit served by those systems. If the project consists of a partial replacement of equipment or ducts (change-out) then load calculations are not required. Select N/A.

~~8.~~11. Enter the total heating load for the dwelling unit described by this document. For projects involving dwelling units with more than one system, this will be a sum of the loads for the parts of the dwelling unit served by those systems. If the project consists of a partial replacement of equipment or ducts (change-out) then load calculations are not required. Select N/A.

~~9.~~12. Enter the number of bedrooms in the dwelling unit

Section B. Design Space Conditioning (SC) System Component Specifications from CF1R

1. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
2. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
3. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
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6. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
7. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.

7b. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R~~LMCC~~), which must be completed prior to this document.

8. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
9. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
10. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
11. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
12. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.

Section C. Installed Space Conditioning (SC) System Component Information

1. Select System name from the list of systems identified in previous sections and originally specified on the CF1R.
2. Briefly describe the area served by this system. Examples: entire house, upstairs, downstairs, sleeping area, north wing, etc.

3. Enter the conditioned floor area served by the system described in this row. The total value of this column for all rows must equal the total dwelling unit conditioned floor area as shown in Section A.
4. This field is filled out automatically. It appears in Section B and is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document, but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
5. This field is filled out automatically. It appears in Section B and is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
6. This field is filled out automatically. It appears in Section B and is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
7. This field is filled out automatically. It appears in Section B and is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
8. This field is filled out automatically. It appears in Section B and is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
9. This field is filled out automatically. It appears in Section B and is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
10. This field is filled out automatically. It appears in Section B and is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
11. If the space conditioning system is a multiple-split system, then enter the number of ducted/ductless indoor units (AHU) connected to the outdoor unit. If the system is a type that does not have an outdoor unit, such as a heating-only type that uses only a furnace air-handling unit, enter 1 for the number of indoor units (The furnace air-handling unit is an indoor unit).

Section D. Installed Heating Equipment Information (not heat pumps)

1. This field is filled out automatically. It is referenced from the same row and column in the previous section.
2. This field is filled out automatically. It is referenced from the same row and column in the previous section.
3. Enter a brief name or description of the indoor unit area served. Examples: Master Bedroom, Dining Room, Living Room, etc

4. If the indoor unit is used to bring outdoor air into the dwelling, the system may be used to comply with the IAQ mechanical ventilation requirements. This is called central fan integrated ventilation (CFI). Systems that have only one indoor unit may use CFI ventilation if yes is selected in this field. Systems with more than one indoor unit connected to one outdoor unit may not select yes.
5. Enter the description of the duct system on this indoor unit. The possible choices are Ductless; Ducted >10ft length, Ducted ≤10ft length.
6. This field is filled out automatically. It is referenced from the same row and column in the previous section
7. Enter the certified heating efficiency of the *installed* equipment. This value is verified against the minimum value shown in Section C. The installed efficiency must be greater than or equal to the required minimum efficiency.
8. Enter the name of the *installed* Heating Unit Manufacturer as shown on the equipment nameplate.
9. Enter the name of the *installed* Heating Unit Model Number as shown on the equipment nameplate.
10. Enter the name of the *installed* Heating Unit Serial number as shown on the equipment nameplate.
11. Enter the rated heating capacity (output) of the *installed* Heating Unit in Btu/h.

Section E. Installed Cooling System Outdoor Unit or Package Unit Equipment Information (not heat pumps).

1. This field is filled out automatically. It is referenced from the same row and column in the previous section.
2. This field is filled out automatically. It is referenced from the same row and column in the previous section.
3. Enter the certified cooling efficiency type for the installed equipment. Select a type from the list provided.
4. Enter the certified cooling efficiency of the *installed* equipment. This value is verified against the minimum value shown in Section B. The installed efficiency must be greater than or equal to the required minimum efficiency.

4b. Enter the certified cooling efficiency of the installed equipment. This value is verified against the minimum value shown in Section B. The installed efficiency must be greater than or equal to the required minimum efficiency.

5. Enter the name of the *installed* Condenser or Package Unit Manufacturer as shown on the equipment nameplate.
6. Enter the name of the *installed* Condenser or Package Unit Model Number as shown on the equipment nameplate.
7. Enter the name of the *installed* Condenser or Package Unit Serial Number as shown on the equipment nameplate.
8. Enter the sensible cooling capacity at design conditions of the *installed* cooling system in Btu/h. This information is found in the system performance information on the manufacturer's published documentation for the installed system.
9. Enter the *installed* Condenser Nominal Cooling Capacity in tons. Note that this is based on the condenser, not the coil or air handler. This can usually be determined by the condenser model number.

Section F. Installed Split System Indoor Coil or Fan Coil Unit Equipment Information (applicable to DX or hydronic heating/cooling coils or fan coil units)

1. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
2. This field is filled out automatically. It is referenced from the same row and column in the previous sections.

3. Enter a brief name or description of the indoor unit area served. Examples: Master Bedroom, Dining Room, Living Room, etc..
4. Enter the type of indoor unit or air handling unit installed by selecting one of the choices from the list.
5. Enter the description of the ducts system on this indoor unit. The possible choices are Ductless; Ducted >10ft length, Ducted ≤10ft length.
6. If the indoor unit is used to bring outdoor air into the dwelling, the system may be used to comply with the IAQ mechanical ventilation requirements. This is called central fan integrated ventilation (CFI). Systems that have only one indoor unit may use CFI ventilation if yes is selected in this field. Systems with more than one indoor unit connected to one outdoor unit may not select yes.
7. Enter the name of the *installed* Indoor Coil or Fan Coil Unit Manufacturer as shown on the equipment nameplate.
8. Enter the name of the *installed* Indoor Coil or Fan Coil Unit Model Number as shown on the equipment nameplate.
9. Enter the name of the *installed* Indoor Coil or Fan Coil Unit Serial Number as shown on the equipment nameplate.
10. If there are multiple indoor units connected to the outdoor unit, enter the nominal cooling capacity (ton) from the nameplate of the indoor unit.

Section G. Installed Heat Pump System – Split System Condensing Unit or Package Unit Equipment Information

1. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
2. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
3. Enter the name of the *installed* Heat Pump Condenser or Package Unit Manufacturer as shown on the equipment nameplate.
4. Enter the name of the *installed* Heat Pump Condenser or Package Unit Model Number as shown on the equipment nameplate.
5. Enter the name of the *installed* Heat Pump Condenser or Package Unit Serial Number as shown on the equipment nameplate.

Section H. Installed Heat Pump System – Efficiency and Performance Compliance Information

1. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
2. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
3. This field is filled out automatically. It is referenced from the same row in Section C.
4. Enter the certified heating efficiency of the *installed* equipment. This value is verified against the minimum value shown in Section C. The installed efficiency must be greater than or equal to the required minimum efficiency.
5. Enter the certified heating capacity at 47°F of the *installed* equipment. This value is verified against the minimum value shown in Section C. The installed capacity must be greater than or equal to the required minimum capacity.
6. Enter the certified heating capacity at 17°F of the *installed* equipment. This value is verified against the minimum value shown in Section C. The installed capacity must be greater than or equal to the required minimum capacity.
7. Enter the certified cooling efficiency of the *installed* equipment. This value is verified against the minimum value shown in Section C. The installed efficiency must be greater than or equal to the required minimum efficiency.

8. Enter the certified cooling efficiency of the *installed* equipment. This value is verified against the minimum value shown in Section C. The installed efficiency must be greater than or equal to the required minimum efficiency.
8b. Enter the certified cooling efficiency of the installed equipment. This value is verified against the minimum value shown in Section C. The installed efficiency must be greater than or equal to the required minimum efficiency.
9. Enter the sensible cooling capacity at design conditions of the *installed* cooling system in Btu/h.
10. Enter the *installed* Condenser Nominal Cooling Capacity in tons. Note that this is based on the condenser, not the coil or air handler. Can usually be determined by the condenser model number.

Section I. Installed Duct System Information

1. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
2. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
3. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
4. This field is filled out automatically. It appears in Section B and C, and is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
5. Enter the R-value of the *installed* supply ducts. This value is verified against the minimum value shown in Section C. The installed R-value must be greater than or equal to the required minimum R-value.
6. This field is filled out automatically. It appears in Section B and C, and is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document. This value may be overwritten in this document but valid discrepancies with the CF1R are uncommon. Overwriting the default value will automatically flag this entry and subject it to additional scrutiny by QA and enforcement personnel.
7. Enter the R-value of the *installed* return ducts. This value is verified against the minimum value shown in Section C. The installed R-value must be greater than or equal to the required minimum R-value.
8. The duct system needs to meet minimum R-6 requirement except for portions of ducts located in conditioned space. Duct systems that are entirely in conditioned space can be uninsulated, subject to ~~HERS~~ECC verification.
9. For newly constructed systems taking the performance credit for better than default air flow or fan efficacy, field verification of these criteria is required and this field is filled out automatically. Otherwise, the user may pick the appropriate choice. Refer to section 150.0(m)13 and Residential Compliance Manual Chapter 4.4 for more information.
10. Specify the number of air filter devices installed in this indoor unit's duct system. Air filter devices installed in completely new systems must be properly sized, as documented in the next section. The value entered here will determine the number of rows needed in the following section.
11. If the system is of a type that can use one of the approved protocols for testing the airflow rate, then enter yes. Otherwise enter no. Note: the protocol in RA3.3.3.1.5 (Alternative to Compliance with Minimum System Airflow Requirements for Altered Systems) is not one of the protocols that is allowed to be used to justify a "yes" to this question.

12. If the system is of a type that can use the approved protocol protocols for verifying the indoor unit's fan efficacy, then answer yes. Otherwise answer no.
13. This field is filled out automatically for some system types. Otherwise select the value that describes the length of the duct system.
14. This field is filled out automatically.

Section J. Installed Air Filter Device Information

1. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
2. This field is filled out automatically. It is referenced from the same row and column in the previous sections
3. This field is filled out automatically. It is referenced from the same row and column in the previous sections
4. Enter a descriptive name of each air filter device so that it may be distinguished from others in the same system. Examples: FG1, filter2, etc.

~~5.~~ Select the appropriate type of filter device from the list.

~~6.~~5.

~~7.~~6. Enter the design flow in CFM of the filter device. The total for all filter devices in a single system should be greater than or equal to the total system design CFM in cooling mode (or heating mode for heat-only systems).

~~8.~~7. Enter the nominal depth of the filter in inches. This is the dimension that is parallel to the airflow. many filters available for sale are 1-inch depth. The ~~2022~~2025 Standards encourage use of 2-inch depth filters.

~~9.~~8. Enter the nominal length of the filter. for example, if the filter is 20" x 30", enter 30.

~~10.~~9. Enter the nominal width of the filter, for example, if the filter is a 20" x 30", enter 20.

~~11.~~10. This field is calculated automatically based on your entries in 8 and 9.

~~12.~~11. This value is calculated automatically for 1-inch depth filters. 2-inch depth or greater filters may use a value determined by the system designer.

~~13.~~12. This field determines whether a 1-inch depth filter complies with the sizing requirements in section 150.0(m)12. A 2-inch depth or greater filter may use the face area determined by the system designer, however most systems have to meet airflow rate and fan efficacy requirements.

~~14.~~13. Enter the design static pressure drop determined by the system designer if 2-inch or greater filters are used. For 1-inch depth filters, the maximum pressure drop is mandatory 0.1 inch W.C.. Filters installed in the filter grille/rack must be capable of meeting this maximum pressure drop at the design airflow rate, as shown on the manufacturer's filter label. Not accounting for higher filter pressure drops will result in poor system airflow characteristics, reduced capacity and reduced efficiency. This may result in not passing field verification.

Section K. Air Filter Device Requirements.

This table is a list of requirements for air filter devices.

Section L. ~~HERS~~ECC Verification Requirements for duct systems

1. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
2. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
3. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
4. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
5. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
6. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
7. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.
8. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.

Section M. ~~HERS~~ECC Verification Requirements for Space Conditioning Equipment

1. This field is filled out automatically. It is referenced from the same row and column in the previous sections.
2. This field is filled out automatically. It is referenced from the same row and column in the previous sections
3. This field is filled out automatically. It is calculated based on data from the CF1R and from previous sections in this document.

Section N. Space Conditioning Systems, Ducts and Fans – Mandatory Requirements and Additional Measures

This table is a list of mandatory measures and additional requirements for space conditioning systems, ducts and fans.

Section O. Test of Defrost Delay Timer Setting (Section 150.0(h)6)

This table is certification requirements for Test of Defrost Delay Timer Setting

Section P. Test of Supplementary Heating Lockout Section 150.0(H)7

This table is certification requirements for Test of Supplementary Heating Lockout

A. General Information					
01	Dwelling Unit Name	<<reference text from CF1R if available, else use the data in the Project Name field>>	02	Climate Zone	<< reference text from CF1R>>
03	Dwelling Unit Total Conditioned Floor Area (ft^2)	<<numeric: xxxxx; **if1 parent is CF1R-PRF**, then if2 project scope = Newly Constructed (Addition Alone) then prompt user to enter a value equal to dwelling unit existing CFA + addition CFA else reference the value from CF1R endif2 **elseif parent is CF1R-NCB-01**, then **if3 project scope = New Addition greater than 1,000 ft2** then prompt user to enter a value equal to dwelling unit existing CFA + addition CFA **elseif project scope = Newly Constructed Building, then** if4 building type = Single Family, then reference value from CF1R-NCB field A10 **elseif parent is CF1R-ADD-01**, then if5 building type= Single Family, then reference value from field A08 from the CF1R-ALT-02 that is required for the dwelling unit according to CF1R-ADD-01 Section J. **elseif parent is CF1R-ALT-01**, then if6 building type= Single Family, then reference value from field A08 from the CF1R-ALT-02 that is required for the dwelling unit according to CF1R-ALT-01 Section G.	04	Number of Space Conditioning Systems in this Dwelling Unit	<<integer: xx; If parent is CF1R-ALT-02 doc type, then use as default the value referenced from CF1R-ALT-02 Section A (field A10); or allow user to override the default and input a new value; flag non-default values and report in project status notes field; elseif parent is not CF1R-ALT-02 doc type, then user input the integer value>>

		elseif parent is CF1R-ALT-02, then reference value from CF1R-ALT-02 field A08. endif1 allow user to override default and input a value; flag overridden values and report in project status notes field >>			
05	Certificate of Compliance Type	<< reference document type property from CF1R: allowed values: performance (CF1R-PRF); or prescriptive additions/alterations (CF1R-ADD/CF1R-ALT); or prescriptive newly constructed (CF1R-NCB)>>	06	Method Used to Calculate HVAC Loads (See Section 150.0(h))	<<user select from list: *ASHRAE Handbook; *SMACNA Residential Comfort System Installation Standards Manual; *ACCA Manual J *n/a equipment changeout, like-for-like *Exception 1 to Section 150.0(h)1: Block loads, (the total load for all rooms combined that are served by the central equipment,) may be used for the purpose of system sizing for additions >>
07	Outdoor Design Condition Source (See Section 150.0(h)2		08	Cooling Outdoor Design Temperature Selected (°F)	
~~07~~ 09	Heating Outdoor Design Temperature Selected (°F)~~Calculated Dwelling Unit Sensible Cooling Load (Btu/h)~~	<<user entry: numeric: xxxxx (allow n/a entry if "n/a equipment changeout, like-for-like" is selected in A06) >>	~~08~~ 10	Calculated Dwelling Unit Sensible Cooling Load (Btu/h) ~~Calculated Dwelling Unit Heating Load (Btu/h)~~	<<user entry: numeric: xxxxx (allow n/a entry if "n/a equipment changeout, like-for-like" is selected in A06) >>
11	Calculated Dwelling Unit Heating Load (Btu/h)		~~09~~ 12	Dwelling Unit Number of Bedrooms	<<<<calculated field: integer xx: if CertComplianceType=performance, then use as default the value referenced from CF1R-PRF or allow user to override the default and input a new value constrained to be greater than or equal to the default value from the CF1R-PRF; flag non-default values and report in project status notes field; elseif parent is not CF1R-PRF doc type, then user input the integer value xx>>
	Determination of Mech01 type (this field not visible to user)	<<calculated field: if1 CertComplianceType=performance, then if2 CF1R-PRF Project Scope=one of the following two types: **Addition and/or Alteration **Newly Constructed - Addition Alone then display doc variation MCH-01d; elseif CF1R-PRF Project Scope=Newly			

		Constructed, then display doc variation MECH01a endif2 elseif CertComplianceType=prescriptive additions/alterations, then display doc variation MECH01b, elseif CertComplianceType=prescriptive newly constructed, then display doc variation MECH01c (this field not visible to user) endif1>>			

Notes:

- The outdoor design temperatures for heating shall be ≥99.0% Heating Dry Bulb or the Heating Winter Median of Extremes values.
- The outdoor design temperatures for cooling shall be ≤1.0% Cooling Dry Bulb and Mean Coincident Wet Bulb values.

MCH-01c - Space Conditioning Systems, Ducts, and Fans - Prescriptive, Newly Constructed Buildings

B. Design Space Conditioning (SC) System Component Specifications from CF1R

This table reports the space conditioning system features that were specified on the registered CF1R compliance document for this project.

<<require one row of data for each SC System identified on the CF1R report that is applicable to this dwelling unit [note: Use section K on CF1R-NCB-01; do not allow user to overwrite these referenced data >>

01	02	03	04	05	06	07	07b	08	09	10	11	12
SC System ID/Name from CF1R	Heating System Type	Heating Efficiency Type	Heating Efficiency Value	Cooling System Type	Cooling Efficiency Type	Cooling Efficiency Value SEER/SEER 2	Cooling Efficiency Value EER/EER2/ CEER	Distributio n System Type	Duct Location	Duct R-value	Thermosta t Type	Comments
<auto filled text: referenced from CF1R>>	<<auto filled text: referenced from CF1R for this system name>>	<<**if** B02 = one of the following: *hydronic, *hydronic + forced air, *combined hydronic, *combined *hydronic + forced air, *hydronic HP *hydronic HP + forced air, **then** value = NA; **else** auto fill text referenced from CF1R for this system name>>	<< **if** B02 = one of the following: *hydronic, *hydronic + forced air, *combined hydronic, *combined hydronic + forced air, *hydronic HP *hydronic HP + forced air, **then** value = NA; **else** auto fill text referenced from CF1R for this system name>>	<<auto fill text referenced from CF1R for this system name>>	auto fill text referenced from CF1R for this system name, **if** one of the following two conditions are true: 1:[data is not available on CF1R] 2:[B05=NoCooling], **then** display result=N/A >>	auto filled text referenced from CF1R for this system name, **if** one of the following two conditions are true: 1:[data is not available on CF1R] 2:[B05=NoCooling], **then** display result=N/A >>	auto filled text referenced from CF1R for this system name, **if** one of the following two conditions are true: 1:[data is not available on CF1R] 2:[B05=NoCooling], **then** display result=N/A >>	<<auto fill text referenced from CF1R for this system name>>	<<auto fill text referenced from CF1R for this system name>>	<<auto fill text referenced from CF1R for this system name if the data is available, **elseif** CF1R does not provide R-value data for this system, **then** result=N/A>>	<<auto fill text referenced from CF1R for this system name, allow N/A>>	<<auto fill text referenced from CF1R for this system name>>

C. Installed Space Conditioning (SC) System Component Information

<< require one row of data to be entered in this table for each of the quantity of space conditioning systems entered in A04.>>

01	02	03	04	05	06	07	08	09	10	11
SC System ID/Name from CF1R	SC System Description of Area Served	Conditioned Floor Area Served by the System (ft^2)	Heating System Type	Cooling System Type	Distribution System Type	Duct Location	SC System Thermostat Type	Cooling Zoning Type	Cooling System Compressor Speed Type	Number of Indoor Units for this System
<< User select applicable system name from a list comprised of the systems identified in column B01; If SC system names from the CF1R are installed more than once in this dwelling unit, then duplicate SC System names are allowed in this field>>	<<user input, text, 15 characters maximum Require each entry to be unique in this dwelling unit i.e. unique within the scope of this instance of the MCH-01>>	<< user input, numeric, xxxx; *Require the sum of the values in this column to be equal to the value in A03 as condition of completion of the doc>>	<< reference value from B02 as default; allow user to override the default and pick one from list: *central gas furnace; *central split HP; *central packaged HP *central large packaged HP *ductless mini-split HP; *room HP; *boiler; *hydronic; *combined hydronic; *hydronic+forced air; *combined hydronic+forced air; *hydronic HP, *hydronic HP+forced air; *gas wall furnace; *gas space heater; *electric ; *Non-Air-Source Heat Pump; *Wood Heat; *Small duct high velocity HP; *ductless VRF HP; *Packaged gas furnace *multisplit HP-ducted *multisplit HP-ductless *multisplit HP-ducted+ductless *VCHP-ducted *VCHP-ductless *VCHP-ducted+ductless *ducted mini-split HP *no heating *SPVHP; *PTHP; **if** value =no heating, **then check**: there must be at least one heating system entered in this section in column C04 to comply, **else** flag noncompliant condition (no heating installed) and do not allow registration to continue. flag non-default values and report in project status notes field; a revised CF1R may be required >>	<< reference value from B05 as default; allow user to override the default and pick one from list: *central split AC; *central split HP *central packaged AC ; *central packaged HP *central large packaged AC ; *central large packaged HP *ductless mini-split AC; *ductless mini-split HP; *gas absorption AC *room AC; *room HP; *Hydronic; *hydronic HP, *hydronic HP+forced air; *evaporative - direct *evaporative - indirect *evaporative - indirectdirect *evaporatively cooled condenser *Ground source heat pump; *Ice Storage AC *no cooling; *Non-air-cooled Air Conditioner; *Non-air-source Heat Pump; *small duct high velocity HP; *small duct high velocity AC; *ductless VRF HP; *ductless VRF AC; *multisplit AC-ducted *multisplit AC-ductless *multisplit AC-ducted+ductless *multisplit HP-ducted *multisplit HP-ductless *multisplit HP-ducted+ductless *VCHP-ducted *VCHP-ductless *VCHP-ducted+ductless *ducted mini-split AC *ducted mini-split HP *SPVAC; *SPVHP; *PTAC; *PTHP; note: cooling system type "**No Cooling"** is the flag for heating-only system type; *flag non-default values and report in project status notes field; a revised CF1R may be required >>	<< reference value from B08 as default. Allow user to override default and pick one from list: ***Ducted** - Air distribution systems with forced air ducts ***DuctsNone** - Air distribution systems without forced air ducts * Multiple split Indoor Units combined Ducted and Ductless >>	<<reference value from B09 as default. Allow user to override default and pick one from list: ***DuctsAttic** - Ducts located overhead in unconditioned attic ***DuctsCrawl** - Ducts located underfloor in unconditioned crawl space ***DuctsGarage** - Ducts located in an unconditioned garage ***DuctsInEx12** - Ducts located within the conditioned space (except < 12 lineal ft) ***DuctsInAll** - HVAC system(s) with all HVAC ducts located in conditioned space ***DuctsOutdoor** - Ducts located in exposed outdoor locations ***DuctsMultiplPL** - Ducts located in multiple places *flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<reference value from B11 as default; allow user to override the default and pick one from list: *setback *OCST per JA5 *Energy Management System >>	<<if cooling system type (C05) = NoCooling, then display text value=n/a; else: user pick one from list: *Zonally Controlled, *NotZonal>>	<<if cooling system type (C05) = NoCooling, then display text value=n/a; else: user pick one from list: *Multi-Speed *Single Speed>>	<< **if** [C04 or C05] = one of the following system types: *room HP *gas wall furnace; *gas space heater; *electric ; *Wood Heat; *Packaged gas furnace *central packaged AC ; *central packaged HP *central large packaged AC ; *central large packaged HP *room AC; *room HP; *evaporative - direct *evaporative - indirect *evaporative - indirectdirect **then** text value=N/A, elseif the CF1R requires use of a Central Fan Integrated (CFI) IAQ Ventilation system, **then** integer value=1, **else** default integer value =1; allow user to overwrite the default to enter one of the following two: 1: an integer value greater than 1, 2: text value=N/A>>

Notes:

D. Installed Heating Equipment Information (not heat pumps)

<<<if all of the SC Systems listed in Section C have a value in C04 = one of the heat pump types (see list that follows below), then display the section does not apply message;
else for each of the SC Systems in section C for which the Heating System Type listed in C04 ≠ one of the heat pump types in the list that follows below; do the following two actions:
1:[require one row of data for each of the SC systems in section C column C02 for which C11=N/A];
2:[for systems for which C11 ≥1, and C04=central gas furnace; require one row of data for each of the quantity of indoor units specified in C11].

*central split HP;
*central packaged HP
*central large packaged HP
*ductless mini-split HP;
*hydronic HP,
*hydronic HP+forced air;
*room HP;
*ducted mini-split HP
*small duct high velocity HP;
*ductless VRF HP
*air-to-water HP
*ground-source HP
*VCHP-Ducted
*VCHP-Ductless
*VCHP-Ducted+Ductless
*multisplit HP-ducted
*multisplit HP-ductless
*multisplit HP-ducted+ductless;
*SPVHP;
*PTHP>>

01	02	03	04	05	06	07	08	09	10	11
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Does Indoor Unit Provide CFI IAQ Ventilation?	Indoor Unit Duct Status	Heating Efficiency Type	Heating Efficiency Value	Heating Unit Manufacturer	Heating Unit Model Number	Heating Unit Serial Number	Rated Heating Capacity Output (Btu/h)
<<auto filled from C01>>	<<auto filled from C02>>	<<**if** value in C11=N/A, **then** value=N/A **elseif** value in C11=1, then value autofilled from C02; **else** user input, text, 15 characters maximum; as default, require value to be unique in this dwelling unit (i.e. unique within the scope of this instance of the MCH-01) except for the case when all the following three conditions are true: 1:[D01=F01] 2:[D02=F02] 3:[C04=central gas furnace] allow user to override the default uniqueness rule if necessary>>	<<if C11 > 1, then value=no; **elseif** IAQ vent system type for this dwelling on the CF1R=one of the following three, *Balanced, *Balanced ERV *Balanced HRV **then** value=no, **elseif** C11=N/A, and C04=Packaged gas furnace, **then** user pick one from list: *Yes *No **else**,user pick one from list: *Yes *No>>	<<**if** the distribution system type value in C06 =DuctsNone, **then** value=Ductless; elseif, C06=Ducted **then** user pick one of the following two values: *Ducted>10ft length *Ducted ≤10ft length **elseif** C06=- *Multiple split Indoor Units Combined Ducted and Ductless, **then** user pick one of the following three text values: *Ductless *Ducted >10ft length *Ducted ≤10ft length>>	<<reference value from B03; allowed values are *AFUE; *COP; *HSPF; *HSPF2; *NA >>	<<**if** B04 = NA, **then** text value in this field = NA; **else** user input numeric value, 100.0≥xx.x≥0.0; check value must be ≥ value in B04, to comply; flag non-compliant values and do not allow registration to proceed >>	<<user input alphanumeric text string max 50 characters>>	<<user input alphanumeric text string max 50 characters>>	<<user input alphanumeric text string max 50 characters>>	<<user input, numeric, xxxx>>

Notes:

E. Installed Cooling System Outdoor Condensing Unit or Package Unit Equipment Information (not heat pumps)

<< if all of the SC Systems listed in Section D have a value in C05= No Cooling, then display the section does not apply message; else require one row of data to be entered in this table for each of the SC Systems listed in Section C for which C05≠no cooling and the Cooling System Type listed in C05 ≠ one of the heat pump types in the following list:

*central split HP;
*central packaged HP
*central large packaged HP
*ducted mini-split HP
*ductless mini-split HP;
*hydronic HP,
*hydronic HP+forced air;
*room HP;
*small duct high velocity HP;
*ductless VRF HP
*air-to-water HP
*ground-source HP
*VCHP-Ducted
*VCHP-Ductless
*VCHP-Ducted+Ductless
*multisplit HP-ducted
*multisplit HP-ductless
*multisplit HP-ducted+ductless;
*SPVHP;
*PTHP>>

01	02	03	04	04b	05	06	07	08	09
SC System ID/Name from CF1R	SC System Description of Area Served	Cooling Efficiency Type	Cooling Efficiency Value SEER/SEER2	Cooling Efficiency Value EER/EER2/CEER	Condenser or Package Unit Manufacturer	Condenser or Package Unit Model Number	Condenser or Package Unit Serial Number	System Cooling Capacity at Design Conditions (Btu/h)	Condenser Nominal Cooling Capacity (tons)
<<auto filled from C01>>	<<auto filled from C02>>	<<as default autofill value from B06 allow user to override the default and pick one value from following list: *EER/SEER; *EER2/SEER2; *CEER; flag non-default values and report in project status notes field; a revised CF1R may be required>>	<<user input, numeric value, xx.x; check value: **if** B07≠N/A, **then** value must be ≥ value in B07, to comply; **elseif** value in B07=N/A and E03=EER/SEER, then value must be ≥14 to comply. Elseif value in B07=N/A and E03=EER2/SEER2, then value must be ≥~~13.4~~14.3. Elseif E03 = CEER, then value = N/A flag non-compliant values and do not allow registration to proceed if not in compliance>>	<<user input, numeric value, xx.x; Check value if B07b≠N/A, then value must be ≥ value in B07b, to comply; elseif value in B07b=N/A and E03=EER/SEER, then value must be ≥ 11.7 to comply; Elseif value in B07b=N/A and E03=EER2/SEER2, then value must be ≥ 11.2; Flag non-compliant values and do not allow registration to proceed if not in compliance>>	<<user input alphanumeric text string max 50 characters>>	<<user input alphanumeric text string max 50 characters>>	<<user input alphanumeric text string max 50 characters>>	<<user input, numeric, xxxxxx>>	<<user input, numeric, x.x>>

Notes:

F. Installed Split System Indoor Unit (Coil or Fan Coil) Equipment Information - applicable to DX or hydronic, heating or cooling, coils and fan coil units.

Systems with more than one indoor coil or fan coil unit (e.g. multi-split systems) shall provide information for each of the system indoor unit coils or fan coil units.

<<<**if** none of the SC Systems listed in Section C have a value in either C04 or C05 = one of the system types in the list that follows below, then display the section does not apply message;
else for each of the SC Systems listed in Section C for which one of the System Types listed in either C04 or C05 is one of the system types in the list that follows below, **require** one row of data to be entered in this table for each of the quantity of indoor units specified in C11 for that system.

*central split AC;
*central split HP
*ductless mini-split AC;
*ductless mini-split HP;
*ducted mini-split HP
*hydronic + forced air;
*combined hydronic + forced air;
*hydronic HP+forced air;
*gas absorption AC;
*ducted mini-split AC
*evaporatively cooled condenser;
*Ice Storage AC;
*small duct high velocity AC;
*small duct high velocity HP;
*ductless VRF AC;
*ductless VRF HP
*air-to-water HP
*ground-source HP
*VCHP-Ducted
*VCHP-Ductless
*VCHP-Ducted+Ductless
*multisplit AC-ducted+ductless
*multisplit AC-ducted
*multisplit AC-ductless
*multisplit HP-ducted
*multisplit HP-ductless
*multisplit HP-ducted+ductless;
>>

01	02	03	04	05	06	07	08	09	10
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Indoor Unit Type	Indoor Unit Duct Status	Does Indoor Unit Provide CFI IAQ Ventilation?	Indoor Unit Manufacturer	Indoor Unit Model Number	Indoor Unit Serial Number	Indoor Unit Nominal Cooling Capacity (tons)
<<auto filled from C01>>	<<auto filled from C02>>	<< **if** value in C11≥ 1, **and** ALL of the following three conditions are true: 1:[D01=F01] 2:[D02=F02] 3:[C04= central gas furnace] **then** value = same value as D03; **elseif** value in C11=1, then value autofilled from C02; **else** user input, text, 15 characters maximum as default, require value to be unique in this dwelling unit (i.e. unique within the scope of this instance of the MCH-01), except for the case when all the following three conditions are true: 1:[D01=F01] 2:[D02=F02] 3:[C04=central gas furnace] allow user to override the default uniqueness rule if necessary>>	<<user pick from list: *HP coil *AC Coil *fancoil AHU *non-furnace airhandler+coil>>	<<if value in C11≥ 1, and ALL of the following three conditions are true: 1:[D01=F01] 2:[D02=F02] 3:[C04=central gas furnace] **then** value = same value as D05; **elseif** the distribution system type value in C06 =DuctsNone, **then** value=Ductless; **elseif**, C06= Ducted, **then** user pick one of the following two values: *Ducted>10ft length *Ducted ≤10ft length **elseif** C06=- *Multiple split Indoor Units Mixed Ducted and Ductless, **then** user pick one of the following three values: *Ductless *Ducted >10ft length *Ducted ≤10ft length>>	<< **if** value in C11≥ 1, **and** ALL of the following three conditions are true: 1:[D01=F01] 2:[D02=F02] 3:[C04= central gas furnace] **then** value = same value as D04; **elseif** C11> 1, **then** value=no; elseif IAQ vent system type for this dwelling on the CF1R= Balanced, then value=no, else, user pick one from list: *Yes *No>>	<<user input alphanumeric text string max 50 characters>>	<<user input alphanumeric text string max 50 characters>>	<<user input alphanumeric text string max 50 characters>>	<<**if** the following three conditions are ALL true: condition 1:[C11 > 1], condition 2:[system type in C04 or C05=one of the following two values: {VCHP-Ducted}, {VCHP - Ducted+Ductless}] condition 3:[F05= one of the following two values: {Ducted >10ft length} {Ducted ≤10ft length}], **then** user input numeric value, x.xx, **else** display text: "value not required">>

Notes:

G. Installed Heat Pump System – Split System Condensing Unit or Package Unit Equipment Information

<<<if none of the SC Systems listed in Section C have a value in C04 or C05 = one of the heat pump types (see list that follows), then display the section does not apply message; else require one row of data to be entered in this table for each of the SC Systems for which the System Type listed in C04 or C05 = one of the heat pump types in the following list:

*central split HP;
*central packaged HP
*central large packaged HP
*ducted mini-split HP
*ductless mini-split HP;
*hydronic HP,
*hydronic HP+forced air;
*room HP
*small duct high velocity HP;
*ductless VRF HP
*air-to-water HP
*ground-source HP
*VCHP-Ducted
*VCHP-Ductless
*VCHP-Ducted+Ductless
*multisplit HP-ducted
*multisplit HP-ductless
*multisplit HP-ducted+ductless;
*SPVHP;
*PTHP>>

01	02	03	04	05
SC System ID/Name from CF1R	SC System Description of Area Served	Condenser or Package Unit Manufacturer	Condenser or Package Unit Model Number	Condenser or Package Unit Serial Number
<<auto filled from C01>>	<<auto filled from C02>>	<<user input alphanumeric text string max 50 characters>>	<<user input alphanumeric text string max 50 characters>>	<<user input alphanumeric text string max 50 characters>>

Notes:

H. Installed Heat Pump System – Efficiency and Performance Compliance Information

<<<if none of the SC Systems listed in Section C have a value in C04 or C05 = one of the heat pump types (see list that follows), then display the section does not apply message; else require one row of data to be entered in this table for each of the SC Systems for which the System Types listed in C04 or C05 = one of the heat pump types in the following list:

*central split HP;
*central packaged HP
*central large packaged HP
*ducted mini-split HP
*ductless mini-split HP;
*hydronic HP,
*hydronic HP+forced air;
*room HP;
*small duct high velocity;
*ductless VRF HP
*air-to-water HP
*ground-source HP
*VCHP-Ducted
*VCHP-Ductless
*VCHP-Ducted+Ductless
*multisplit HP-ducted
*multisplit HP-ductless
*multisplit HP-ducted+ductless;
*SPVHP;
*PTHP>>

01	02	03	04	05	06	07	08	08b	09	10
SC System ID/Name from CF1R	SC System Description of Area Served	Heating Efficiency Type	Heating Efficiency Value	System Rated Heating Capacity at 47°F (Btu/h)	System Rated Heating Capacity at 17°F (Btu/h)	System Cooling Efficiency Type	System Rated Cooling Efficiency Value SEER/SEER2	System Rated Cooling Efficiency Value EER/EER2/CEER	System Cooling Capacity at Design Conditions (Btu/h)	Condenser Nominal Cooling Capacity (ton)
<<auto filled from C01>>	<<auto filled from C02>>	<<reference value from B03; allowed values are: *AFUE; *COP; *HSPF; *HSPF2; or *NA>>	<<**if** B04 = NA, then text value = NA; **else** user input, numeric value, 100.0≥xx.x≥0.0; check value: if B04≠N/A, then value must be ≥ value in B04, to comply; flag non-compliant value and do not allow registration to proceed>>	<<user input, numeric value, xxxxx.x>>	<<user input, numeric value, xxxxx.x if available, else allow user to pick text value= N/A >>	<<as default reference value from B06. allow user to override the default and pick one value from following list: *EER/SEER; *EER2/SEER2; *CEER; flag non-default values and report in project status notes field; a revised CF1R may be required>>	<< user input, numeric value, 100.0≥ xx.x≥0.0; check value: if B07≠N/A, then value must be ≥ value in B07, to comply; elseif value in B07=N/A and E03=SEER, then value must be ≥14 to comply; elseif vale in B07=N/A and E03=EER2/SEER2, then value must be ≥~~13.4~~14.3. Elseif H07 = CEER, then value = N/A flag non-compliant values and do not allow registration to	<<user input, numeric value, xx.x; check value: if B07b≠N/A, then value must be ≥ value in B07b, to comply; elseif value in B07b=N/A and E03=SEER, then value must be ≥11.7 to comply; elseif vale in B07b=N/A and E03=EER2/SEER2, then value must be ≥11.2. flag non-compliant values and do not allow registration to proceed if not in compliance>>	<<user input, numeric, xxxxxx>>	<<user input, numeric, x.x>>

							proceed if not in compliance>>			
	Notes:									

I. Installed Duct System Information

<<**if** all of the SC Systems listed in Section C have a Distribution System Type value in C06 =DuctsNone - Air distribution systems without forced air ducts , **then** display the section does not apply message;

else require one row of data in this table for each space conditioning system in section D field D02 that meets BOTH of the following two conditions: 1:[C04=Packaged Gas Furnace] 2:[C06=ducted]
Also require one row in this table for each indoor unit in section D field D03 that meets BOTH of the following two conditions: 1:[value in C04=central gas furnace], 2:[the value in D03 ≠ F03]
Also require one row in this table for each space conditioning system in section E field E02 that meets ALL of the following three conditions: 1:[value in C05=central packaged AC, or central large packaged AC], 2:[the same packaged unit is not already listed in section D thus D02≠E02]; 3:[value in C06=Ducted]
Also require one row in this table for each space conditioning system in Section G field G02 that meets ALL the following three conditions: 1:[value in C05=central packaged HP, or central large packaged HP], 2:[the same packaged unit is not already listed in section D thus D02≠G02]; 3:[value in C06=Ducted]
ALSO require one row of data in this table for each indoor unit in F03 for which the value in F05 = one of the following two values [*Ducted>10ft length; *Ducted ≤10ft length] >>

01	02	03	04	05	06	07	08	09	10	11	12	13	14
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Supply Duct Location	Supply Duct R-Value	Return Duct Location	Return Duct R-Value	Exception from Min R-Value for Ducts In Conditioned Space	Method of compliance with Airflow and Fan Efficacy Req's in 150.0(m)13	Number of Air Filter Devices on Duct System	Can Approved Airflow Protocols be used to test this System?	Can Approved Fan Efficacy Protocol be used to test this System?	Total Duct Length	Required New Duct R-Value
<<auto filled C01>>	<<auto filled from C02>>	<<**if** system type in C04=[Packaged Gas Furnace], **then** value auto filled from C02, **elseif** system type in [C04 or C05] = one of the following four types: 1: central packaged AC ; 2: central packaged HP 3: central large packaged AC ; 4: central large packaged HP **then** value auto filled from C02, **else** reference applicable values from D03 and F03; >>	<<reference value from C07 as default. Allow user to overwrite and pick one from list: ***DuctsAttic** - Ducts located overhead in unconditioned attic ***DuctsCrawl** - Ducts located underfloor in unconditioned crawl space ***DuctsGarage** - Ducts located in an unconditioned garage ***DuctsInEx12** - Ducts located within the conditioned space (except < 12 lineal ft) ***DuctsInAll** - HVAC system(s) with all HVAC ducts located in conditioned space ***DuctsOutdoor** - Ducts located in exposed outdoor locations ***DuctsMultipIPL** - Ducts located in multiple places	<<user pick from list: *R-0, *R-2.1, *R-4.2 *R-6, *R-8, *R-10, *R-12; **and** check value: must be ≥ value in I14] to comply subject to the following exception: **if** I08= *Ducts <R-6 entirely in Conditioned Space, **then** value <-R-6 complies; else flag non-compliant value and do not allow registration to proceed >>	<<reference value from C07 as default. Allow user to overwrite and pick one from list: ***DuctsAttic** - Ducts located overhead in unconditioned attic ***DuctsCrawl** - Ducts located underfloor in unconditioned crawl space ***DuctsGarage** - Ducts located in an unconditioned garage ***DuctsInEx12** - Ducts located within the conditioned space (except < 12 lineal ft) ***DuctsInAll** - HVAC system(s) with all HVAC ducts located in conditioned space ***DuctsOutdoor** - Ducts located in exposed outdoor locations ***DuctsMultipIPL** - Ducts located in multiple places flag non-default values and report in project status notes field; a revised CF1R may be required >>	<<user pick from list: *R-0, *R-2.1, *R-4.2 *R-6, *R-8, *R-10, *R-12; check value: must be ≥ value in I14 to comply subject to the following exception: **if** I08= *Ducts <R-6 entirely in Conditioned Space, **then** value <R-6 complies; else flag non-compliant value and do not allow registration to proceed >>	<< Default Value=No Exceptions, allow user to override the default and select one or more of the following two values: *portions of uninsulated or <R-6 ducts in conditioned space; **if** values in both I04 and I06= *DuctsInAll-conditioned space-entirely, **then** also allow user to select the following value: *Ducts uninsulated or <R-6 ducts entirely in conditioned space>>	<< **if** System Type in C05=no cooling, **then** text value = "Exempt - No Cooling"; **elseif** System Type in C05=one of the following three system types: *evaporative - direct, or *evaporative - indirect, or *evaporative - indirectdirect, **then** text value = "Exempt - Evaporative System"; **elseif** Section I [field 11 or field 12] =no, **then** text value = "Exempt - Approved Protocols N/A"; **elseif** one or more of the following three conditions is true: **C09=Zonally Controlled **D04=yes (is CFI IAQ System) **F06=yes (is CFI IAQ System) then text value = "~~HERS~~ECC Verified Fan Efficacy and Airflow Rate"; **else** user select one from the following two text values: **"~~HERS~~ECC Verified Fan Efficacy and Airflow Rate;	<<user enter integer value (this value will determine number of rows per indoor unit in next section if the total duct length is greater than 10 ft)>>	<< **if** system type in C04 or C05 is one of the following four system types: *multisplit HP-ducted *multisplit HP-ducted+ductless *multisplit AC-ducted *multisplit AC-ducted+ductless, **then** value=no, **elseif** system type in C04 or C05 is one of the following system types: *central split AC; *central split HP *central packaged AC ; *central packaged HP *central large packaged AC *central large packaged HP, **then** value=Yes, **else** user pick one of the following two values from list: **yes **no **check**:	<< **if** system type in C04 or C05 is one of the following four system types: *multisplit HP-ducted *multisplit HP-ducted+ductless *multisplit AC-ducted *multisplit AC-ducted+ductless, **then** value=no, **elseif** system type in C04 or C05 is one of the following system types: *central split AC; *central split HP *central packaged AC ; *central packaged HP *central large packaged AC *central large packaged HP, **then** value=Yes, **else** user pick one of the following two values from list: **yes **no, **check**: **if** value=no, **then** report in project status notes	<<**if** there is an applicable value in [either D05 or F05]= [Ducted>10ft length], **then** value=[>10ft] **elseif** there is an applicable value in [either D05 or F05]= [Ducted ≤10ft length], **then** value=[≤10ft] **else** user pick one text value from the following 2: *[>10ft] *[≤10ft]>>	<<**if** value in B10≠N/A, **then** reference value from B10 **elseif** if A02= CZ 3, 5-7 **then** value = R-6.; **elseif** A02=CZ 1, 2, 4, 8-16 **then** value = R-8>>

			flag non-default values and report in project status notes field; a revised CF1R may be required >>					**"~~HERS~~ECC verified Return Duct Design per Table 150.0-B, C"; >>		**if** value=no, **then** report in project status notes field that exemption from mandatory ~~HERS~~ECC verification of system airflow has been claimed. Enforcement agency confirmation recommended. >>	field that exemption from mandatory ~~HERS~~ECC verification of Fan Efficacy has been claimed. Enforcement agency confirmation recommended.		

Notes:

J. Installed Air Filter Device Information

Mandatory requirements for air filter devices are specified Section 150.0(m)12. The installer shall place a sticker in or near each filter grille that displays the design airflow rate for that filter grille/rack and the maximum allowed clean filter pressure drop at the design airflow rate. This will inform the occupant of the airflow vs pressure drop performance required for replacement air filters.

<<**if** all of the SC Systems listed in Section C have a Distribution System Type value in C06 =DuctsNone - Air distribution systems without forced air ducts, **then** display the section does not apply message;
elseif there are no duct systems in section I for which I13=[>10ft], then display the section does not apply message;
else require one row of data (each) for the quantity of Air filter devices in Section I field 10 for each of the duct systems listed in Section I for which I13=[>10ft]>>

01	02	03	04	05	06	07	08	09	10	11	12	13
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	Air Filter Name or Description of Location	Air Filter Rack Type	Design Airflow Rate for Air Filter Device (cfm)	Air Filter Nominal Depth (inch)	Air Filter Nominal Length (inch)	Air Filter Nominal Width (inch)	Air Filter Calculated Nominal Face Area ($inch^2$)	Air Filter Required Minimum Face Area ($inch^2$)	Face Area Complianc e	Design Allowable Pressure Drop for Air Filter Device (inch W.C.)
<<auto filled from C01>>	<<auto filled from C02>>	<<auto filled from Section I field 03	<<user input text, maximum 20 characters>>	<<user select from list: *Filter Grille * Mounted at air handler * Mounted in line between return grille and air handler >>	<<user enter numeric value, xxx>>	<<user enter integer value ≥1.00 >>	<<user enter integer value ≥1.00 >>	<<user enter integer value ≥1.00 >>	<<calculated value= J08*J09 >>	<<**if** J07=1, **then** calculated value=(J06 ÷ 150) * 144; **else** display text: "specified by system designer">>	<<**if** value in J11= "specified by system designer", **then** display text: "specified by system designer"; **elseif** J10≥J11, then display text: "complies", else display text:"does not comply">>	<< if value in J07=1, then value = 0.1; else user enter value, numeric, 1.5≥x.xx≥0.01>>

Notes:

K. Air Filter Device Requirements

<<if section J does not apply, then display the section does not apply message; else display this section>>

Mandatory Air Filter Device Requirements can be found in Section 150.0(m)12A-E. Some mandatory requirements may apply in addition to those listed below.

01	All recirculated air and all outdoor air (including make up air) supplied to the occupiable space is filtered before passing through the system's thermal conditioning components.
02	The space conditioning system shall be designed to accommodate the clean-filter pressure drop imposed by the system air filter device(s). The design airflow rate and maximum allowable clean-filter pressure drop at the design airflow rate applicable to each air filter device shall be determined by the system designer. The system installer shall affix a sticker/label to each system air filter grille/rack locations that discloses the filter's design airflow rate and the filter's maximum allowable clean-filter pressure drop at the design airflow rate. The sticker/label shall be permanently affixed to the air filter device, readily legible, and visible to a person replacing the air filter.
03	All system air filter devices shall be located and installed in such a manner as to allow access and regular service by the system owner.
04	The system shall be provided with air filters having a designated efficiency equal to or greater than MERV 13 when tested in accordance with ASHRAE Standard 52.2, or a particle size efficiency rating equal to or greater than 50% in the 0.30-1.0 µm range and equal to or greater than 85 percent in the 1.0-3.0 µm range when tested in accordance with AHRI Standard 680.
05	The system shall be provided with air filters that have been labeled by the manufacturer to disclose efficiency and pressure drop ratings that conform to the efficiency and pressure drop requirements for the air filter grilles/racks.
06	Filter racks or grilles shall use gaskets, sealing, or other means to close gaps around inserted filters and prevent air from bypassing the filter.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

L. ~~HERS~~ECC Verification Requirements for Duct Systems

<<if all of the SC Systems listed in Section C have a Distribution System Type value in C06 =DuctsNone - Air distribution systems without ducts , then display the section does not apply message; **else** require one row of data in this table for each of the indoor units listed in Section I field 03>>

01	02	03	04	05	06	07	08
SC System ID/Name from CF1R	SC System Description of Area Served	Indoor Unit Name or Description of Area Served	MCH-20 Duct Leakage Test	MCH-21 Duct Location Verification	MCH-22 AHU Fan Efficacy (W/cfm)	MCH-23 AHU Airflow Rate (cfm/ton)	MCH-28 Return Duct Design - Table 150.0-B or C
<<auto filled from Section I field 01>>	<<auto filled from Section I field 02>>	<<reference applicable values from Section I field 03>>	<<value=yes>>	<<**if** the value in I08 (Section I field 08)= *Ducts uninsulated or <R-6 ducts entirely in conditioned space, **then** display result in this field=yes; **else** display result=no>>	<<**if** Section I field 12=no, **then** result=no elseif section I field 09 result is "~~HERS~~ECC Verified Fan Efficacy and Airflow Rate", then result is yes, **elseif** C05=no cooling; **AND** one or more of the following two are true: *[D04=yes] (is a CFI system) *[F06=yes] (is a CFI system) **then** result=yes **else** result=no	<<if section I field 11 = no, then result = no; else**if** section I field 09 result is "~~HERS~~ECC Verified Fan Efficacy and Airflow Rate", **then** result = yes, **elseif** the value in M03=yes, **AND** the value in L08=no, **then** result in this field=yes **elseif** C05=no cooling; **AND** one or more of the following two are true: *[D04=yes] (is a CFI system) *[F06=yes] (is a CFI system) **then** result=yes **else** result=no	<< <<**if** Section I field 09 result is: "~~HERS~~ECC Verified Return Duct Design per Table 150.0-B, C"; **then** result=yes **else** result=no>>

Notes:

M. ~~HERS~~ECC Verification Requirements for Space Conditioning Equipment
<<require one row of data in this table for each of the SC Systems listed in E01 and H01>>

01	02	03
SC System ID or Name from CF1R	SC System Description of Area Served	MCH-25 Refrigerant Charge
<<auto filled from C01>>	<<auto filled from C02>>	<<calculated field: if A02= one of the following values: 2, 8, 9, 10, 11, 12, 13, 14, 15; then result=yes, else result=no>>

Notes:

N. Space Conditioning Systems, Ducts and Fans – Mandatory Requirements and Additional Measures
Additional mandatory requirements from Section 150.0 that are not listed here may be applicable to some systems. These requirements may be applicable to only newly installed equipment or portions of the system that are altered. Existing equipment may be exempt from these requirements.

Heating Equipment	
01	**Equipment Efficiency:** All heating equipment must meet the minimum efficiency requirements of Section 110.1 and Section 110.2(a) and the Appliance Efficiency Regulations.
02	**Controls:** All unitary heating systems, including heat pumps, must be controlled by a setback thermostat. These thermostats must be capable of allowing the occupant to program the temperature set points for at least four different periods in 24 hours. See Sections 150.0(i), 110.2(~~b~~c).
03	**Sizing:** Heating load calculations must be done on portions of the building served by new heating systems to prevent inadvertent undersizing or oversizing. See sections 150.0(h)1 and 2
04	**Furnace Temperature Rise:** Central forced-air heating furnace installations must be configured to operate at or below the furnace manufacturer's maximum inlet-to-outlet temperature rise specification. See Section 150.0(h)4.
05	**Standby Losses and Pilot Lights:** Fan-type central furnaces may not have a continuously burning pilot light. Section 110.5 and Section 110.2(d).
Cooling Equipment	
06	**Equipment Efficiency:** All cooling equipment must meet the minimum efficiency requirements of Section 110.1 and Section 110.2(a) and the Appliance Efficiency Regulations.
07	**Refrigerant Line Insulation:** All refrigerant line insulation in split system air conditioners and heat pumps must meet the R-value and protection requirements of Section 150.0(j)1~~2~~ and 2~~3~~, and Section 150.0(m)9.
08	**Condensing Unit Location:** Condensing units shall not be placed within 5 feet of a dryer vent outlet. See Section 150.0(h)3A.
09	**Liquid Line Filter Drier:** A liquid line filter drier shall be installed according to the manufacturer's specifications 150.0(h)3B.
10	**Sizing:** Cooling load calculations must be done on portions of the building served by new cooling systems to prevent inadvertent undersizing or oversizing. See Section 150.0(h)1 and 2.
Cooling and Heating Equipment (Other requirements):	
11	**System Selection:** See section 150.0(h)5 A. Equipment sizing and selection shall meet the cooling and heating loads of Section 150.0(h)1 and 2. B. Systems shall be sized based on ACCA Manual S-2023 in accordance with these requirements: iv. **Cooling Capacity:** There is no limit on the minimum capacity. v. **Furnaces**: Heating capacity shall be sized based on ACCA Manual S-2023, Table N2.5. vi. **Heat Pump Heating Capacity**:

	a. Minimum: Heating systems are required to have a heating capacity meeting the minimum requirements of the CBC not including any supplementary heating. b. Maximum: There is no limit on the maximum heating capacity.
12	**Defrost:** See section 150.0(h)6 and the exceptions. A. If a heat pump is equipped with an installer adjustable defrost delay timer, the delay timer shall be set to greater than or equal to 90 minutes. B. The installer shall certify on the Certificate of Installation (CF2R) that the control configuration has been tested in accordance with the testing procedure in the CF2R. **Exception 1 to Section 150.0(h)6.** Dwelling units in Climate Zones 6 and 7. **Exception 2 to Section 150.0(h)6.** Dwelling units with a conditioned floor area of 500 square feet or less in Climate Zones 3, 5 through 10, and 15.
13	**Supplementary heating control configuration:** See section 150.0(h)7. Heat pumps with supplementary heat, including, but not limited to, electric resistance heaters or gas furnace supplementary heating, shall comply with the following requirements: **See section 150.0(h)7 for exceptions.** A. Lock out supplementary heating above an outdoor air temperature of no greater than 35°F. There are additional thermostat requirements in section 150.0(i)2. B. The installer shall certify on the Certificate of Installation that the control configuration has been tested in accordance with the testing procedure found in the CF2R. C. The controls may allow supplementary heater operation above 35°F only during defrost; or when the user selects emergency operation.
14	**Sizing of electric resistance supplementary heat:** See section 150.0(h)8. For heat pumps with electric resistance heating, the capacity of electric resistance heat shall not exceed the heat pump nominal cooling capacity (at 95°F ambient conditions) multiplied by 2.7 kW per ton, rounded up to the closest kW.
15	**Capacity variation with third-party thermostats:** See section 150.0(h)9 Variable or multi-speed systems shall comply with the following requirements: A. The space conditioning system and thermostat together shall be capable of responding to heating and cooling loads by modulating system compressor speed, and meet thermostat requirements in section 150.0(i)2.
Air Distribution System Ducts, Plenums and Fans	
~~11~~ 16	**Insulation:** The minimum duct insulation value is R-6 or ducts can be uninsulated if the duct system is located entirely in conditioned space. Note that higher values may be required by the prescriptive or performance requirements. See Section 150.0(m)1B for exceptions.
~~12~~ 17	**Connections and Closures:** All installed air-distribution system ducts and plenums must meet the requirements of CMC Sections 601.0, 602.0, 603.0, 604.0, 605.0 and ANSI/SMACNA-006-2006.
Heat Pump Thermostat	
~~13~~ 18	A thermostat shall be installed that meets the requirements of Section 110.2(b) and Section 110.2(c).
~~14~~ 19	The thermostat shall be installed in accordance with the manufacturers published installation specifications.
~~15~~ 20	First stage of heating shall be assigned to heat pump heating.
~~16~~ 21	Second stage back up heating shall be set to come on only when the indoor set temperature cannot be met.
22	Setback thermostats: All heating or cooling systems, including heat pumps, not controlled by a central energy management control system (EMCS) shall have a setback thermostat, as specified in Section 110.2(c). See Section 150.0(i)1
23	Thermostats that are applied to heat pumps with supplemental heating: See Section 150.0(i)2 The thermostats controlling heat pumps with electric resistance supplementary heat or gas furnace supplementary heat shall comply with the following requirements: See Section 150.0(i)2 for exceptions. A. The thermostat shall receive outdoor air temperature from an outdoor air temperature sensor or from an internet weather service. B. The thermostat shall display the outdoor air temperature. C. The thermostat and heat pump shall lock out supplementary heat when the outdoor air temperature is above 35°F. D. The thermostat shall have an indicator to notify when supplementary heat or emergency heat is in use. E. During defrost or when the user selects emergency heating, supplementary heat operation is permitted above 35°F.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

O. Test of Defrost Delay Timer Setting (Section 150.0(h)6) The installing contractor shall confirm that a heat pump's installer-adjustable Defrost Delay Timer Setting (if it exists) is set to no less than 90 minutes.		
01	Test Applicability. Select the statement describing test applicability for this project: 1. The test applies because the heat pump utilizes an installer adjustable Defrost Delay Timer Setting to control defrost and there are no exceptions. 2. The test does not apply because the heat pump does not utilize an installer-adjustable Defrost Delay Timer Setting to control defrost. 3. The test does not apply because Exception 1. Dwelling units in Climate Zones 6 and 7 applies. 4. The test does not apply because Exception 2 **.** Dwelling units with a conditioned floor area of 500 square feet or less in Climate Zones 3, 5 through 10, and 15 applies.	<<user selects from list: 1. if test applies and no exceptions; 2. If test does not apply because no adjustable Defrost Delay Timer Setting; 3. If test does not apply because of Exception 1; 4. If test does not apply because of Exception 2; 5. N/A>>
<<If O01 = 1. (Applies with no exceptions), then display fields 02-06 for completion Else If O01 = 2. (No adjustable timer setting), then only display the static text "This test in not applicable because there is no installer adjustable Defrost Delay Timer Setting to control defrost" Else If O01 = 3 (Exception 1) Then only display the static text "This test in not applicable because of Exception 1"; Else If O01 = 4 (Exception 2) Then only display the static text "This test in not applicable because of Exception 2"; Else If O01 = 5. N/A Then do not display anything else >>>>		
02	Recording Configuration of Controls. Specify the mechanism for setting the Defrost Delay Timer Setting (for example, the name of defrost delay timer setting in the thermostat setup, or the location and number of the specific dip switch, jumper, or dial that adjusts Defrost Delay timer).	<<user Input: Text >>
03	Record the heat pump's Maximum Available Defrost Delay Timer Setting (minutes).	<< user Input: User Input: IntegerNonnegative>>
04	Record where you set the Defrost Delay Timer Setting (fo example, the numeric timer setting, dip switch position, jumper configuration, or dial setting).	<< user Input: Text>>
05	Record where you set the Defrost Delay Timer Setting, in minutes.	<< user Input: IntegerNonnegative >>
06	Confirming Configuration of Controls. If possible, the Defrost Delay Timer Setting must be 90 minutes or greater. Confirm the Defrost Delay Timer Setting is at least 90 minutes.	<< user selects from list: Yes, or No - Do Not Proceed >>

P. Test of Supplementary Heating Lockout		
The installing contractor shall confirm the heat pump or thermostat is configured to lock out supplementary heating anytime the outdoor air temperature is above 35°F.		
01	Test Applicability. Select the statement describing test applicability for this project: 1. The test applies because the heating system is a heat pump with supplementary heating and there are no exceptions. 2. The test does not apply because the heating system does not include a heat pump with auxiliary heating. 3. The test does not apply because Exception 1. Heat pump being installed in a single family dwelling unit in Climate Zones 7 or 15. 1. The test does not apply because Exception 2 . Heat pump being installed in a single family dwelling unit with a conditioned floor area of 500 square feet or less.	<<user selects from list: 1. if test applies and no exceptions; 2. If test does not apply because heating system does not include a heat pump with auxiliary heating; 3. If test does not apply because of Exception 1; 4. If test does not apply because of Exception 2; 5. N/A >>
<<If P01 = 1. (Applies with no exceptions), then display fields 02-06 for completion Else If P01 = 2. (the heating system does not include a heat pump with auxiliary heating), then only display the static text “This test in not applicable because there the heating system does not include a heat pump with auxiliary heating” Else If P01 = 3 (Exception 1) Then only display the static text “This test in not applicable because of Exception 1”; Else If P01 = 4 (Exception 2) Then only display the static text “This test in not applicable because of Exception 2”; Else If P01 = 5. N/A Then do not display anything else >>>>		
02	If supplementary heating is electric resistance, what is its rated capacity (kW)?	<< user input, decimalnonnegative; else N/A >>
03	If supplementary heating is gas, what is its rated input capacity (kBtu/hr)?	<< user input, numeric, xxxxx; else N/A >>
04	Confirming Configuration of Controls. Specify the mechanism for locking out the supplementary heating (for example, the name of supplementary heating lockout setting in the thermostat setup, or the location and number of the specific dip switch, jumper, or dial that adjusts lockout temperature). If there is no mechanism to lock out supplementary heating above a temperature no greater than 35°F this test fails.	<< user Input: Text >>
05	Record supplementary heating lock-out setting (for example, the numeric thermostat lock-out setpoint, dip switch position, jumper configuration, or dial setting).	<< user input: Text >>
06	At what Outdoor Air Temperature (OAT) are the controls configured to begin locking out supplementary heating? If this number is above 35°F this test fails.	<< user entry: numeric, x.x degF; else if Q06 value > 35 degF, Do Not Proceed >>